



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 10	Team No- 27	Project Title- Online Learning Management System (LMS)
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1. Abstract

The Online Learning Management System (LMS) is a full-stack e-learning platform designed to bridge the gap between instructors and learners by offering an integrated environment for course delivery, assessment, and progress tracking. Traditional classroom-only learning struggles to scale, track engagement, and provide learners with flexible, verifiable outcomes, while many existing LMS tools are either too rigid or lack real-time interactivity. This project addresses that gap by building a complete platform where instructors can author and publish courses, upload and stream video lectures, and create quizzes and assignments, while students can enroll, learn at their own pace, attempt assessments, and track their progress toward certification. The proposed solution uses React.js for a responsive front-end, Node.js and Express for a scalable back-end API, and MySQL for structured, relational storage of users, courses, and progress data. AWS S3 is used for reliable, scalable storage and streaming of video lecture content, JWT ensures secure, role-based authentication for students, instructors, and admins, and Socket.io enables real-time features such as live discussion forums and notifications. Core modules include User and Role Management, a Course Builder, Video Lecture Streaming, a Quiz and Assignment Engine, Progress Tracking and Certificate generation, a Discussion Forum, and an Admin Analytics dashboard for monitoring platform-wide engagement and performance.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090026	Venkat Siddharth Reddy Moku	Siddharth. M
2520090055	Sai Saketh Vutharavalli	Sai Saketh V
2520090028	Dhanush Reddy	K. R. D.

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Front-end: React.js. Back-end: Node.js, Express.js. Database: MySQL. Authentication: JWT (JSON Web Tokens). Cloud Storage: AWS S3 (for video lecture hosting and streaming). Real-time Communication: Socket.io (for discussion forum and live notifications).

Guide Name and Signature (remarks if any)